

Auteur: Thijs Van Schel
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$$z^4 + 4(1 - i)z^2 + 3 - 4i = 0$$

$$\text{Stel } t = z^2$$

$$D = (4 - 4i)^2 - 4 \cdot (3 - 4i)$$

$$= 16 - 32i - 16 - 12 + 16i = -12 - 16i$$

$$\lambda^2 + 12\lambda - 64 = 0$$

$$D = 12^2 - 4 \cdot (-64) = 400$$

$$\lambda_1 = \frac{-12 + 20}{2} = 4 \Rightarrow t_1 = \pm 2$$

$$\lambda_2 = \frac{-12 - 20}{2} = -16 \Rightarrow t_2 = \pm 4$$

$$\Rightarrow \{2 - 4i, -2 + 4i\}$$

$$z_1 = \frac{-4 + 4i + 2 - 4i}{2} = -1$$

$$z_2 = \frac{-4 + 4i - 2 + 4i}{2} = -3 + 4i$$

$$t = -1 \Rightarrow z_{1,2} = \pm i$$

$$t = -3 + 4i \Rightarrow z^2 = -3 + 4i$$

$$\begin{cases} x^2 - y^2 = -3 \\ 2xy = 4 \\ x^2 + y^2 = \sqrt{9 + 16} = 5 \end{cases}$$

$$2x^2 = 2 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$$2y^2 = 8 \Rightarrow y = \pm 2$$

$$z_3 = 1 + 2i$$

$$z_4 = -1 - 2i$$

$$\Rightarrow V = \{i, -i, 1 + 2i, -1 - 2i\}$$

References